

NOUADHIBOU, Mauritania

WMO: 614150

Lat: 20.920N

Lon: 17.036W

Elev: 1

StdP: 101.32

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	14.7	-0.2	3.7	21.7	1.8	4.3	21.5	12.4	17.5	11.6	17.1	7.3	0	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.8	34.0	21.2	32.1	21.1	30.8	21.1	25.5	28.7	24.5	28.0	23.6	27.4	5.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.7	27.0	23.2	18.0	26.3	22.3	17.0	25.7	78.0	29.5	73.9	28.3	70.5	27.7	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.4	10.7	11.6	39.2	1.9	1.4	10.2	40.2	9.0	41.1	7.9	41.9	6.5	42.9		
(5)			7.4	26.6	1.5	1.7	6.3	27.8	5.4	28.7	4.6	29.7	3.5	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	20.3	20.5	21.3	21.4	22.1	23.4	23.7	25.0	26.4	25.5	23.4	21.4	(6)
(7)		DBStd	2.82	1.82	2.06	2.24	2.00	1.99	1.92	1.70	1.98	2.28	2.59	2.23	1.74	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	7	3	3	1	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD10.0	4701	319	295	351	341	374	403	426	466	492	479	403	353	(10)
(11)		CDD18.3	1667	64	64	93	91	116	153	167	207	242	221	153	96	(11)
(12)		CDH23.3	7538	222	249	341	249	356	523	567	1122	1637	1299	668	306	(12)
(13)	CDH26.7	2101	36	58	107	66	91	119	106	282	529	460	199	49	(13)	
(14)	Wind	WSAvg	6.8	6.1	6.9	7.0	7.6	8.1	8.1	7.2	6.4	6.5	6.3	5.8	5.4	(14)
(15)	Precipitation	PrecAvg	17	2	2	1	1	0	0	1	2	4	3	1	0	(15)
(16)		PrecMax	49	17	41	14	15	5	3	10	17	30	74	20	8	(16)
(17)		PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	14	3	7	3	2	1	0	2	4	7	11	4	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	30.9	33.0	32.8	33.2	33.2	32.0	33.3	36.8	36.8	33.8	29.9	(19)
(20)			MCWB	16.7	16.5	17.1	18.7	19.6	20.7	22.1	24.2	22.6	21.6	19.1	16.9	(20)
(21)		2%	DB	27.2	28.4	29.8	28.2	28.9	29.8	29.1	31.1	33.1	33.8	31.1	28.0	(21)
(22)			MCWB	15.4	16.0	16.6	18.3	18.9	20.5	22.1	23.8	23.1	21.0	18.2	16.4	(22)
(23)		5%	DB	25.8	26.4	27.0	25.9	26.5	27.7	27.5	29.6	31.1	31.2	29.0	26.6	(23)
(24)			MCWB	15.0	15.9	17.3	18.1	18.7	20.4	21.9	23.6	23.2	21.0	18.1	16.3	(24)
(25)		10%	DB	24.1	24.3	24.8	24.1	25.0	26.1	26.1	28.0	29.7	29.0	27.0	25.0	(25)
(26)			MCWB	15.1	16.0	17.2	18.2	18.8	20.0	21.5	23.0	23.1	20.7	18.6	16.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	20.8	20.6	21.2	21.8	23.5	25.7	26.9	27.0	25.0	22.7	21.9	(27)
(28)			MCD B	23.1	24.0	25.6	25.8	27.0	26.9	27.0	29.0	29.3	29.4	25.6	23.7	(28)
(29)		2%	WB	19.1	19.6	19.6	20.2	20.6	22.1	23.7	25.3	25.8	23.3	21.4	20.6	(29)
(30)			MCD B	22.4	22.8	23.9	24.2	25.2	26.1	26.6	28.6	28.9	28.2	25.3	23.0	(30)
(31)		5%	WB	18.1	18.6	19.0	19.5	20.0	21.2	22.6	24.4	24.6	22.5	20.6	19.5	(31)
(32)			MCD B	21.6	22.0	22.6	23.3	24.3	25.2	25.8	27.7	28.3	27.6	24.6	22.7	(32)
(33)		10%	WB	17.2	17.7	18.4	18.9	19.4	20.7	21.8	23.6	23.7	21.9	20.0	18.5	(33)
(34)			MCD B	20.9	21.2	22.0	22.6	23.4	24.6	25.2	27.0	28.0	26.7	23.9	22.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.8	9.9	9.7	8.7	8.6	7.0	6.9	8.8	9.3	9.4	9.1	(35)	
(36)			MCDBR	11.7	12.2	12.5	11.0	10.9	10.5	9.1	9.0	10.7	11.9	11.3	10.9	(36)
(37)			MCWBR	5.5	5.3	4.7	4.0	3.9	3.6	3.5	3.8	4.0	4.1	4.1	4.7	(37)
(38)		5% WB	MCD BR	9.7	9.6	9.9	9.1	9.1	8.7	7.4	8.0	8.5	9.5	9.2	8.2	(38)
(39)			MCWBR	5.0	4.9	4.2	3.9	3.9	4.0	4.0	4.3	4.8	4.1	4.1	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.444	0.438	0.484	0.493	0.555	0.701	0.789	0.777	0.724	0.561	0.438	0.437	(40)
(41)		taud		2.027	2.065	1.953	1.977	1.834	1.593	1.477	1.522	1.592	1.865	2.140	2.093	(41)
(42)		Ebn at Noon		815	851	827	823	766	656	601	610	633	729	818	807	(42)
(43)		Edh at Noon		159	162	188	186	213	269	302	290	267	196	141	144	(43)
(44)	All-Sky Solar Radiation	RadAvg		4.95	5.88	6.92	7.62	7.93	7.77	7.36	7.01	6.44	5.80	5.03	4.47	(44)
(45)		RadStd		0.27	0.31	0.28	0.33	0.27	0.32	0.27	0.19	0.26	0.32	0.33	0.28	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page